Note

In the above derivation, we took the case where rays are converged to a point after the refraction. But in general, the rays can either converge or diverge after refraction. This depends upon the value of n_2 over n_1, radius of curvature R and position of object u. The other possibility is discussed in example 6.12.

EXAMPLE 6.12

Find the position of the image of a point object O in the two cases given. Take the radius of curvature of the surface R as 15 cm, $n_1 = 1$ and $n_2 = 2$.

Case i) O is located 10 cm to the left of the surface.

Case ii) O is located 30 cm to the left of the surface.

Solution

Case i)

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R};$$

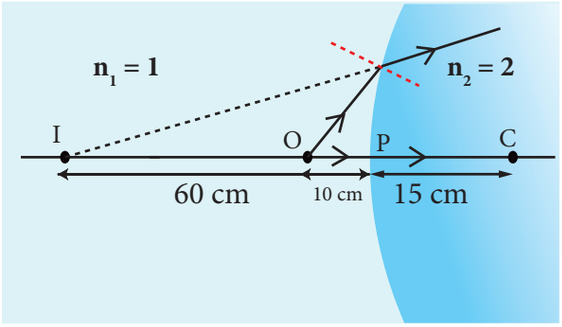
applying sign convention,

u = –10 cm, R = 15 cm

$$\frac{2}{v} - \frac{1}{(-10)} = \frac{(2-1)}{15}; \frac{2}{v} = \frac{1}{15} - \frac{1}{10}$$

$$\therefore v = -60 \text{ cm}$$

[a virtual image is formed 60 cm, to the left of the surface]



Case ii)

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R};$$

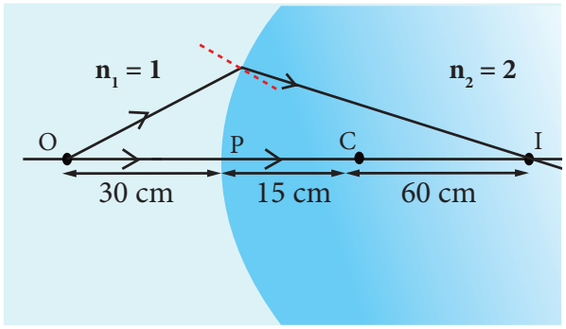
applying sign convention,

u = –30 cm, R = 15 cm

$$\frac{2}{v} - \frac{1}{(-30)} = \frac{(2-1)}{15}; \frac{2}{v} + \frac{1}{30} = \frac{1}{15}$$

$$\therefore v = 60 \text{ cm}$$

[a real image is formed 60 cm, to the right of the surface]



6.6 THIN LENS

A lens is formed by a transparent material bounded between two spherical surfaces (or) one plane and another spherical surface. In a thin lens, the distance between the surfaces is very small. For two spherical surfaces, there will be two centres of curvature C_1 and C_2 and correspondingly two radii of curvature R_1 and R_2. For a plane surface, the centre of curvature C is at infinity and its radius of curvature R is infinity ($R = \infty$). The terminologies of spherical mirrors hold good very much for thin lenses except for the focal length.

6.6.1 Primary and secondary focus

As the thin lens is formed by two surfaces, the lens may separate two different media. i.e. the media to the left and right of

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